

Question 4 – Concept Deep-Dive

Assigned Topic: Why 100×100 Face Images Live in a Low-Dimensional Subspace (Manifold Hypothesis)

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Intuition

A 100×100 grayscale face image is represented by 10,000 numbers, so each sample is a point in $\mathbb{R}^{10,000}$. The key idea is that this is an *ambient* dimension, not the true complexity of the face-generation process. Valid face images are heavily constrained: facial parts appear in structured locations, neighboring pixels are correlated, and variation is mostly controlled by a few latent factors (identity, illumination, pose, and expression). So even though coordinates are high-dimensional, real face data occupy a much smaller region of that space. This is exactly the intuition emphasized in dimensionality-reduction lectures: data can be high-dimensional in representation but low-dimensional in intrinsic structure [1, 4, 2].

Mathematical Core

Let $x_i \in \mathbb{R}^D$ with $D = 10,000$ and $X \in \mathbb{R}^{N \times D}$ be mean-centered face data. PCA looks for a unit direction u that maximizes projected variance:

$$\max_{u: u^T u = 1} u^T \Sigma u, \quad \Sigma = \frac{1}{N} X^T X.$$

An equivalent reconstruction form is minimizing

$$\sum_{i=1}^N \|x_i - uu^T x_i\|^2,$$

which gives the same eigenvector solution (top eigenvector of Σ). For d' dimensions, keep top eigenvectors $v_1, \dots, v_{d'}$.

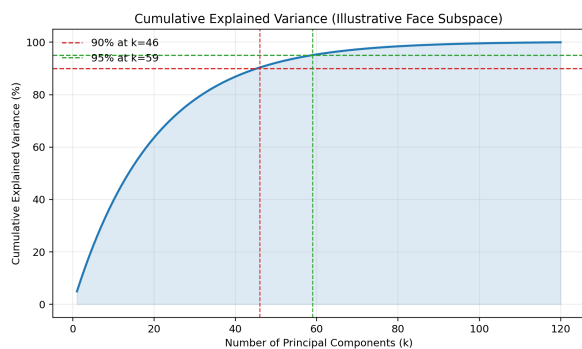
Any mean-centered face can be expanded in eigen-basis form:

$$x = \sum_{j=1}^D (x^T v_j) v_j = \sum_{j=1}^{d'} \alpha_j v_j + \sum_{j=d'+1}^D \alpha_j v_j,$$

where truncating to first d' terms gives the eigenface approximation

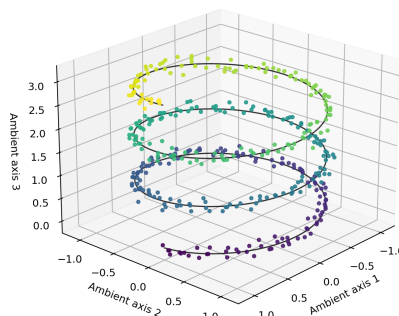
$$\hat{x} = \sum_{j=1}^{d'} \alpha_j v_j, \quad \alpha_j = x^T v_j.$$

This matches the low-rank approximation view: retain major variation and discard low-energy directions. With SVD $X = UDV^T$, PCA keeps only leading components (rank- d' approximation), so faces can be represented with far fewer than 10,000 coefficients [1, 2, 5, 4]. The manifold hypothesis adds that PCA gives a local linear view of a globally nonlinear face manifold [3, 6].



(a) CEV rises rapidly: a few components explain most variance.

Illustration: Low-Dimensional Face Manifold in High-D Space



(b) Low-dimensional manifold embedded in a high-dimensional space.

Figure 1: Visual summary: the subspace and manifold viewpoints are consistent, not contradictory.

In practice, intrinsic dimension is often estimated by choosing the smallest d' such that cumulative explained variance exceeds a threshold (for example, 90% or 95%). For face images, this empirical d' is typically far smaller than 10,000, which is concrete evidence that the data occupy a structured low-dimensional subset rather than the full ambient space.

Connections to Other SMAI Concepts

This topic connects directly to the curse of dimensionality: raw Euclidean distances in pixel space are noisy and often not identity-faithful. Projecting to a compact representation (for example, PCA/eigenfaces) reduces nuisance variation and makes neighborhood geometry more reliable for k-NN. It also links to Fisherfaces/LDA: PCA captures major variation, then LDA uses labels to extract discriminative directions. In short, low-dimensional structure is the bridge between representation, generalization, and classification quality in face-recognition pipelines [1, 2, 4].

The manifold viewpoint also explains why nonlinear dimensionality-reduction methods are useful: when geometry is curved, linear PCA can need many components. Also, low-dimensional does not mean easy; occlusion, pose changes, and misalignment can still break local neighborhoods, and maximizing variance need not maximize separability [6].

References

- [1] SMAI Spring 2026 Lecture Slides, L08–L13: low-rank data, Bayesian decision, MLE/GMMs, probabilistic algorithms, SVMs, and principles of learning.
- [2] C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
- [4] scikit-learn Documentation, [PCA User Guide](#).
- [5] M. Turk and A. Pentland, “Eigenfaces for Recognition,” *Journal of Cognitive Neuroscience*, 3(1), 1991.
- [6] J. B. Tenenbaum, V. de Silva, and J. C. Langford, “A Global Geometric Framework for Nonlinear Dimensionality Reduction,” *Science*, 290(5500), 2000.